



PROSTHETIC ARM USING EMG SENSOR

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Abstract

To Design a Prosthetic Arm Using EMG (Electromyography) to help disabled People.

The only goal is to integrate into society and improve the quality of life of persons with disabilities.

The prosthetic hand is a device used for the amputee to make them capable of doing daily life stuff.

Hardware And Software Requirements

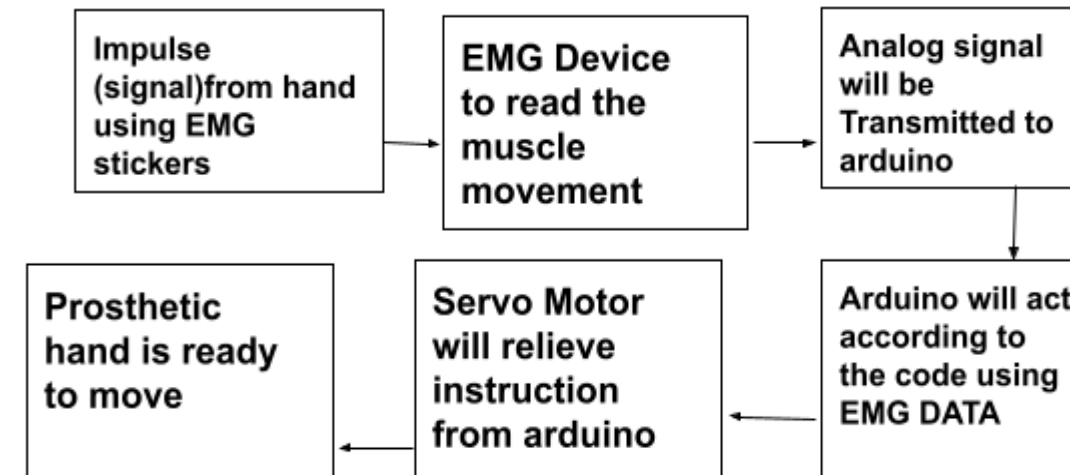
Arduino Board

Servo Motor

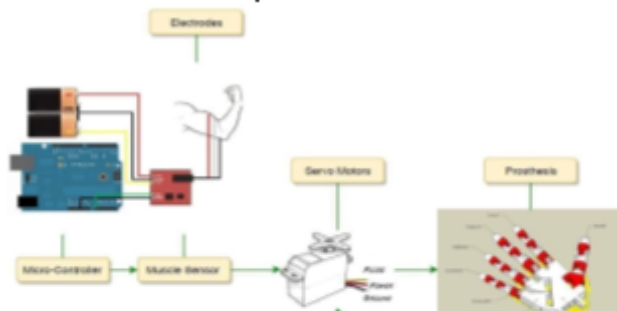
EMG Sensor With Electrodes

Prosthetic Arm

System Description



Implementation



1.This in movable hand is controlled through the sensors connected to the remaining muscle of the Patient's arm

2.Sensors pick up Electromyographic signals which are given to the arduino and converted to movements of the prosthetic arm.

3.Here the servo motors are used to control the fingers movement of the artificial 3D designed and printed hand.

Results

Research into producing a prosthetic hand that can be easily operated from the arm by the EMG signal is also a big step towards prosthetics.

The aim of this project to develop a low-cost, 3D prosthetic hand-printed using EMG.

Though the final device has been fulfilled most of the initial steps, there are other aspects of the new layout that need to be rebuilt for real- life use cases.

For many people, an artificial limb can improve mobility and the ability to manage daily activities, as well as provide the means to stay independent.

Future Scope

FUTURE SCOPE 1:

After successfully designing and testing this project there is some recommendations for future work on this project. First of all, the feedback sensing ability should be improved. To improve the ability of sensing it is very important to increase the ability of the hand to sense in any part of the hand or fingers.

FUTURE SCOPE 2:

On the other hand, this design can tell the amputee if the object touches the finger hardly or softly. To make this project very useful for blind people it is important to activate some feedback which tells the amputee about these properties.

References

IoT Book: The Internet of Things" by Samuel Greengard, Precision: Principles, Practices, and Solutions for the Internet of Things" by Timothy.

S. Hasan, et. al., "Wearable Mind Thoughts Controlled Open Source 3D Printed Arm with Embedded Sensor Feedback System", CHIRA 2018 - 2nd International Conference on Computer-Human Interaction Research and Applications, 2018.